# MATHEMATICAL PRELIMINARIES:

## INTRODUCTION TO FORMAL PROOF:

**Proof:-** A proof is a convencing argument that some statement is true.

*** Deductive Proof :- (Direct Proof):**

- Sequence of statements whose truth leads us from some initial statements called hypothesis to a conclusion statement.

**Ex:**

Prove that of $x \geq 4$ then $2^x \geq x^2$.

**Proof:-** when $x = 4$ then

$$2^x = 2^4 = 16$$
$$x^2 = 4^2 = 16$$
$$\Rightarrow 2^x = x^2.$$

As $x$ grows larger than 4, $2^x$ doubles each time $x$ increase of one.

$\therefore$ Each time $x$ increases above 4, $2^x$ grows more than $x^2$.

Hence Proved.

## ADDITIONAL FORM OF PROOFS:

*** Proof by Contrapositive :-**

- The contrapositive of a statement if H then C is if not C then not H.

To prove a statement it is enough to prove contrapositive.

**Ex:**

Prove that for any integer i, j and n if $i * j = n$ then either $i \leq \sqrt{n}$ or $j \leq \sqrt{n}$.

**Proof:-** The given statement if $i * j = n$ then either $i \leq \sqrt{n}$ or $j \leq \sqrt{n}$.

The contrapositive statement of given is,

$i > \sqrt{n}$ and $j > \sqrt{n}$ then $i * j \neq n$

Let us prove, contrapositive statement

$i > \sqrt{n}$ — ①, $j > \sqrt{n}$ — ②

① $\Rightarrow i > \sqrt{n}$ Multiply both sides by j

$i * j > \sqrt{n} * j$ — ③

② $\Rightarrow j > \sqrt{n}$ Multiply both sides by $\sqrt{n}$

$\sqrt{n} \times j > \sqrt{n} \times \sqrt{n}$, $\sqrt{n} \times j > n$ — ④

From ③ & ④, $i * j > \sqrt{n} * j > n$

$i * j \Rightarrow n$, $i * j \neq n$.

The contrapositive of given statement is true. Hence the given statement also true.

## INDUCTIVE PROOFS:

### MATHEMATICAL INDUCTION :-

$P(n)$ is a statement involving an integer n, to show $P(n)$ is true for all $K \geq n_0$. This proof needs

1. $P(n_0)$ is true
2. If $P(k)$ is true, then $P(K+1)$ is true for $K \geq n_0$.

**Ex:**

S.T $1+2+3+\cdots+n = \frac{n(n+1)}{2}$.

**Step1: Basis:**

If $n=1 \Rightarrow \frac{n(n+1)}{2} = \frac{1(1+1)}{2} = 1$

Hence the proof.

**Step2: Induction:**

Assumption is,

$1+2+3+\cdots+K = \frac{K(K+1)}{2}$ is true.

To prove, $1+2+3+\cdots+(K+1) = \frac{(K+1)(K+2)}{2}$ is true

L.H.S $\underbrace{1+2+3+\cdots+K}+(K+1)$

$= \frac{K(K+1)}{2} + (K+1) = \frac{(K+1)(K+2)}{2}$

Hence the proof. = R.H.S

# CENTRAL CONCEPT OF AUTOMATA THEORY

## FINITE AUTOMATA :-

A Finite Automata is formally denoted by five tuple,

$(Q, \Sigma, \delta, q_0, F)$ where

$Q$ is the set of States
$\Sigma$ is the Input Alphabet
$\delta$ is the transistion function
$q_0$ is Initial State
$F$ is final set of states.

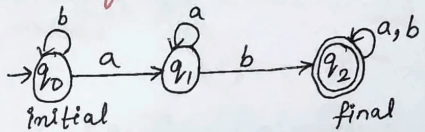
## Types of finite Automata :-

1. Deterministic Finite Automata (DFA)
2. Non-Deterministic Finite Automata (NFA)

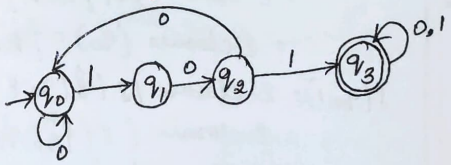
| DFA | NFA |
|---|---|
| 1. From each state for each input symbol there is exactly one transistion. | 1. For each state for each input symbol we can have 0 or more transistions. |
| 2. No $\epsilon$-transistions | 2. $\epsilon$-transistions are allowed. |

## DFA Problems

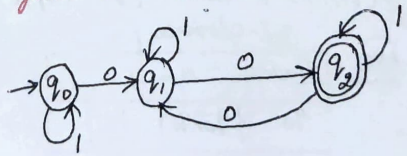
**Ex:1**
Design a DFA for the language having strings with ab as a substring over $\Sigma = \{a, b\}$.




**Ex:2**
Design a DFA for the language having strings with 101 as a substring over $\Sigma = \{0, 1\}$.

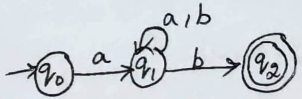



**Ex:3**
Design a DFA for the language having strings with even no of zero's over $\Sigma = \{0, 1\}$.




## NFA Problems.

**Ex:1)**
Design a NFA to accept strings that start with A and end with b over $\Sigma = \{a, b\}$ also write formula def. of NFA. check whether the string abaab is accepted or not!




$\{Q, \Sigma, \delta, q_0, F\}$,
$\{(q_0, q_1, q_2), \{a, b\}, \delta, q_0, q_2\}$

where $\delta$ is,

$\delta(q_0, a) = q_1$ , $\delta(q_0, b) = \phi$
$\delta(q_1, a) = q_1$ $\delta(q_2, a) = \phi$
$\delta(q_1, b) = q_1$
$\delta(q_1, b) = q_2$

String : abaab

$\delta(q_0, abaab) = \delta(q_1, baab)$
$= \delta(q_1, aab)$
$= \delta(q_1, ab)$
$= \delta(q_1, b)$
$= q_2$ //

# EQUIVALENCE OF NFA & DFA:

## CONVERSION NFA to DFA :-

Ex: Construct a DFA equivalent to NFA given below,

$M = (\{q_0, q_1\}, \{0,1\}, \delta, q_0, \{q_1\})$

where $\delta(q_0,0) = \{q_0, q_1\}$

$\delta(q_0,1) = \{q_1\}$

$\delta(q_1,0) = \phi$

$\delta(q_1,1) = \{q_0, q_1\}$

Sol:

Let M1 be the DFA, Let $[q_0]$ be the initial state of DFA, Let $\delta'$ be the transition function of DFA,

$\delta'([q_0],0) = \delta(q_0,0) = \{q_0,q_1\}$

$\delta'([q_0],1) = \delta[q_0,1) = \{q_1\}$

$\delta'([q_0,q_1],0) = \delta(q_0,0) \cup \delta(q_1,0) = \{q_0,q_1\}$

$\delta'([q_0,q_1],1) = \delta(q_0,1) \cup \delta(q_1,1) = \{q_0,q_1\}$

$\delta'([q_1],0) = \delta(q_1,0) = \phi$

$\delta'([q_1],1) = \delta(q_1,1) = \{q_0,q_1\}$.

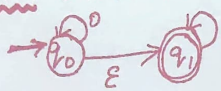
ANS:

| States | 0 | 1 |
|---|---|---|
| $\{q_0\}$ | $\{q_0,q_1\}$ | $\{q_1\}$ |
| $\{q_0,q_1\}$ | $\{q_0,q_1\}$ | $\{q_0,q_1\}$ |
| $\{q_1\}$ | $\phi$ | $\{q_0,q_1\}$ |

# FINITE AUTOMATA with Epsilon Transitions:

## EQUIVALENCE OF NFA with Epsilon Transitions :-

Ex: Construct NFA from NFA with Epsilon,




$\varepsilon$-closure$(q_0) = \{q_0, q_1\}$

$\varepsilon$-closure$(q_1) = \{q_1\}$.

Processing of $q_0$:

$\delta(q_0,0) = \varepsilon$-closure $(\delta(\hat{\delta}(q_0,\varepsilon),0))$

$= \varepsilon$-closure $(\delta(\{q_0,q_1\},0))$

$= \varepsilon$-closure $(q_0) = \{q_0,q_1\}$.

$\delta(q_0,1) = \varepsilon$-closure $(\delta(\hat{\delta}(q_0,\varepsilon),1))$

$= \varepsilon$-closure $(\delta(\{q_0,q_1\},1))$

$= \varepsilon$-closure $(q_1) = \{q_1\}$

Processing of $q_1$:

$\delta(q_1,0) = \varepsilon$-closure $(\delta(\hat{\delta}(q_1,\varepsilon),0))$

$= \varepsilon$-closure $(q_0) = \{q_0,q_1\}$.

$\delta(q_1,1) = \varepsilon$-closure $(\delta(\hat{\delta}(q_1,\varepsilon),1))$

$= \varepsilon$-closure $(q_1) = \{q_1\}$.

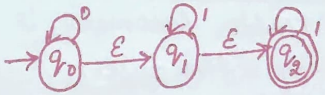
ANS:

| States | 0 | 1 |
|---|---|---|
| $q_0$ | $\{q_0,q_1\}$ | $\{q_1\}$ |
| $q_1$ | $\{q_0,q_1\}$ | $\{q_1\}$ |

⇓ NFA without $\varepsilon$-Transitions.

# CONVERSION OF NFA-$\varepsilon$ to DFA

Ex: Convert the NFA-$\varepsilon$ move giv below to an Equivalent DFA.




$\varepsilon$-closure of $q_0 = \{q_0, q_1, q_2\}$

$\varepsilon$-closure of $q_1 = \{q_1, q_2\}$

$\varepsilon$-closure of $q_2 = \{q_2\}$.

$\varepsilon$-closure of $\{q_0\} = \{q_0,q_1,q_2\} \rightarrow$ (A)

1. $\hat{\delta}(A,0) = \varepsilon$-closure of $(\delta(A,0))$

$= \varepsilon$-closure $(\delta(\{q_0,q_1,q_2\},0))$

$= \varepsilon$-closure $(q_0) = \{q_0,q_1,q_2\}$

$\hat{\delta}(A,1) = \varepsilon$-closure $(\delta(A,1))$

$= \varepsilon$-closure $(q_1) = \{q_1,q_2\} \rightarrow$ (B)

$\hat{\delta}(A,2) = \varepsilon$-closure $(\delta(A,2)) = \varepsilon$-closure

$= \{q_2\} \rightarrow$ (C)

2. $\hat{\delta}(B,0) = \epsilon$-closure $(\delta(B,0))$

$= \epsilon$-closure $(\delta(\{q_1,q_2\},0)) = \phi$.

$\hat{\delta}(B,1) = \epsilon$-closure $(\delta(B,1)) = \varepsilon$-closure

$= \{q_1,q_2\} \rightarrow$ (B)

$\hat{\delta}(B,2) = \epsilon$-closure $(\delta(B,2)) = \varepsilon$-closure

$= \{q_2\} \rightarrow$ (C)

3. $\hat{\delta}(C,0) = \epsilon$-closure $(\delta(C,0)) = \phi$

$\hat{\delta}(C,1) = \phi$.

$\hat{\delta}(C,2) = \varepsilon$-closure $(q_2) = \{q_2\} \rightarrow$ (C).

ANS:

| States | 0 | 1 | 2 |
|---|---|---|---|
| A | A | B | C |
| B | $\phi$ | B | C |
| C | $\phi$ | $\phi$ | C |

# REGULAR LANGUAGE

## Regular Expression

Following Languages by Regular Expression.

Set of Strings of a's and b's of length two

$$(a+b)(a+b)$$

Set Containing Zero or More 0's followed by Single 1

$$(0+0^*1)$$

Set of all strings ending With aba

$$(a+b)^* aba$$

Set over {1} having odd length of String

$$1(11)^*$$

The Set of all Strings 0's is divisible by five is.

$$1^*(00000)^*1^*$$

Set of all strings abb as Substring

$$(a+b)^* abb (a+b)^*$$

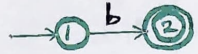
String ends with 1 and does not Contains the Substring 00

$$(1+10)^* (101+1)^* 1$$

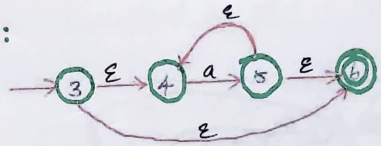
# Equivalence of Finite Automata and Regular Expression

Construction of $\varepsilon$-NFA from the Regular Expression : $b+ba^*$

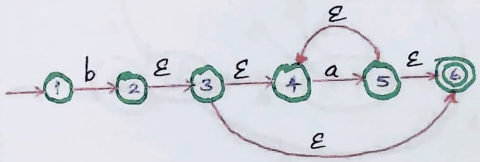
b:



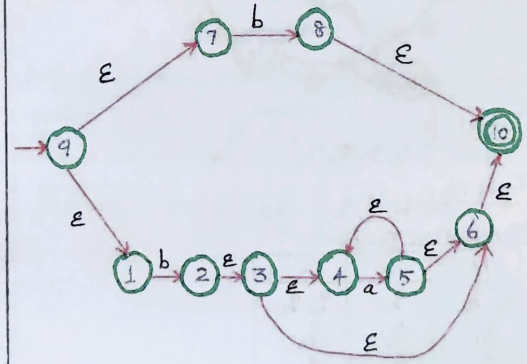

$a^*$:



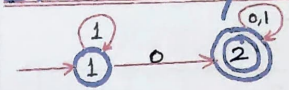

$ba^*$:




$b+ba^*$:




# DFA TO REGULAR EXPRESSION




Let $k=0$,

$$R_{11}^{(0)} = \varepsilon+1$$
$$R_{12}^{(0)} = 0$$
$$R_{21}^{(0)} = \phi$$
$$R_{22}^{(0)} = \varepsilon+0+1$$

IF $k=1$,

$$R_{ij}^{(k)} = R_{ij}^{k-1} + R_{ik}^{k-1}\left(R_{kk}^{k-1}\right)^* \cdot R_{kj}^{(k-1)}$$

$$R_{11}^{(1)} = (\varepsilon+1)^*$$
$$R_{12}^{(1)} = 1^*0$$
$$R_{21}^{(1)} = \phi$$
$$R_{22}^{(1)} = \varepsilon+0+1$$

IF $k=2$,

$$R_{12}^{(2)} = R_{12}^{(1)} + R_{12}^{(1)}\left(R_{22}^{(1)}\right)^* R_{22}^{(1)}$$

$$= 1^*0 + 1^*0\,(\varepsilon+0+1)^*(\varepsilon+0+1)$$

$$= 1^*0 + 1^*0\,(\varepsilon+0+1)^*$$

$$R_{12}^{(2)} = 1^*0\,(\varepsilon+0+1)^*$$

$$\boxed{R_{12}^{(2)} = 1^*0\,(0+1)^*}$$

# MINIMIZATION OF DFA

1)

| States | a | b |
|---|---|---|
| → $q_0$ | $q_1$ | $q_6$ |
| $q_1$ | $q_6$ | $q_2$ |
| $q_2$ | $q_3$ | $q_1$ |
| $q_3$ | $q_3$ | $q_0$ |
| $q_4$ | $q_3$ | $q_5$ |
| $q_5$ | $q_6$ | $q_4$ |
| $q_6$ | $q_5$ | $q_6$ |
| $q_7$ | $q_6$ | $q_3$ |

Input : a

$\{q_0, q_1, q_2, q_4, q_5, q_6, q_7\}$ $\{q_3\}$

Input : b

$\{q_0, q_6\}$ $\{q_1, q_5\}$ $\{q_7\}$ $\{q_2, q_4\}$ $\{q_3\}$

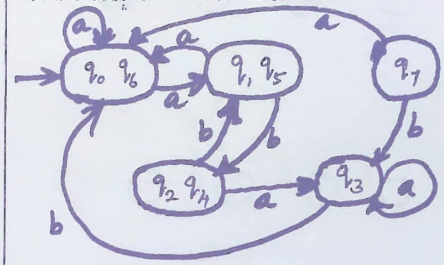
$\{q_0, q_1, q_5, q_6, q_7\}$ $\{q_2, q_4\}$ $\{q_3\}$

$\{q_0 q_6\}$ $\{q_1, q_5\}$ $\{q_7\}$ $\{q_2 q_4\}$ $\{q_3\}$

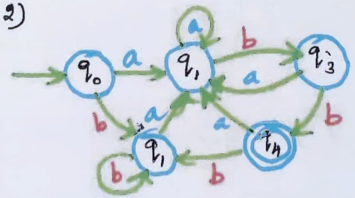
## TRANSITION TABLE

| | a | b |
|---|---|---|
| $q_0 q_6$ | $q_1 q_5$ | $q_0 q_6$ |
| $q_1 q_5$ | $q_0 q_6$ | $q_2 q_4$ |
| $q_7$ | $q_0 q_6$ | $q_3$ |
| $q_2 q_4$ | $q_2$ | $q_1 q_5$ |
| $q_3$ | $q_3$ | $q_0 q_6$ |

## TRANSITION DIAGRAM :




2)




Input : b

$\{q_0, q_1, q_2 q_3\}$ $\{q_4\}$

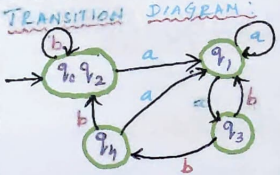
Input : b

$\{q_0, q_1, q_2\}$ $\{q_3\}$ $\{q_4\}$

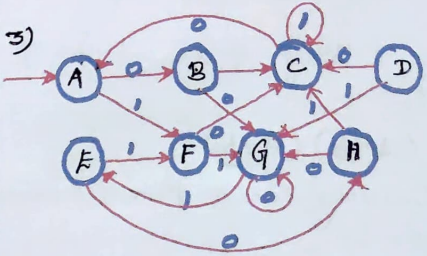
TRANSITION TABLE

| STATE | a | b |
|---|---|---|
| $q_0 q_2$ | $q_1$ | $q_0 q_2$ |
| $q_1$ | $q_1$ | $q_3$ |
| $q_3$ | $q_1$ | $q_4$ |
| $q_4$ | $q_1$ | $q_0 q_2$ |

## TRANSITION DIAGRAM :




3)




Input : 0

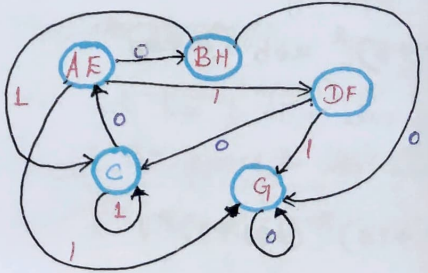
$\{A\ B\ D\ E\ F\ G\ H\}$ $\{C\}$

Input : 1

$\{A\ B\ E\ G\ H\}$ $\{B\ H\}$ $\{D\ F\}$ $\{C\}$

$\{A\ E\}$ $\{G\}$ $\{B\ H\}$ $\{D\ F\}$ $\{C\}$

## TRANSITION DIAGRAM :



## PUMPING LEMMA FOR REGULAR SET:

### PUMPING LEMMA:

Let: $L \rightarrow$ Regular Language

$n \rightarrow$ Constant

Such that for Every string in 'W' in 'L'

Such that $|W| \geq n$

We can break W into three

* $y \neq \varepsilon$
* $|xy| \leq n$
* $xy^kz \in L \; \forall \; k \geq 0$

Note: Repeating $y$ any no of times

Deleting $y$ (or) keeps the resulting string in the Same Language.

### Problem Based on Pumping Lemma:

Prove that the set

$L = \{ 0i^2 \mid i \text{ is an integer}, i \geq 1 \}$

Which consist of all strings of 0's whose length is perfect Square is not regular

Assume the given Language $L$ is a regular

Let us take the Sample string $W = 0^{n^2}$

Where $n$ is the Constant of pumping lemma

By pumping lemma:

We can write, $0^{n^2} = xyz$

Where

1) $y \neq \varepsilon$
2) $|xy| < n$
3) $xy^kz \in L \; \forall \; k \geq 0$

put $k=2$, We get the string $xy^2z$

By pumping lemma,

$xy^2z \in L$

$|xy^2z|$ Must be a perfect Square

Let us find $|xy^2z|$

$|x^2y^2z| = |xyz| + |y|$

$\leq n^2 + n$ $\quad \because |xy| \leq n$ and $y \neq \varepsilon$

$|xy^2z| > n^2 \rightarrow (1)$

$|xy^2z| < (n+1)^2 \rightarrow (2)$

From (1) and (2)

$|xy^2z|$ lies properly b/w two perfect Squares and hence $|xy^2z|$ is not a perfect Square.

$\therefore xy^2z \in L$ which is a Contradiction that difference

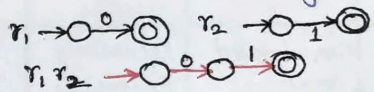
Given Language is not regular.

## CLOSURE PROPERTIES OF REGULAR LANGUAGES

1) Union of two regular Language is Regular.

$L(m) = L(m_1) \cup L(m_2)$

2) Concatenation of two regular Language is Regular.




3) Intersection of two regular Langage is Regular

$L \cap M$ is also regular

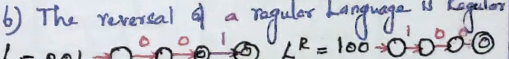
4) The difference of two regular Language is Regular

$L - m$ is Regular $\quad L - m = L \cap \overline{m}$ Regular.

5) The Compliment of Regular Language is Regular

$\overline{L} = \Sigma^* - L$

6) The reversal of a regular Language is Regular

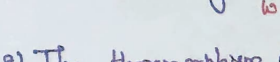



$L^R$ Regular.

7) Closure of Regular in Regular.

$w = a$ $\quad w = a^n$

8) The Homomorphism of Regular Language is Regular




Hence $h(L)$ also Regular.

9) The Inverse homomorphism of a regular Language is Regular.

## GRAMMAR INTRODUCTION:

## TYPES OF GRAMMAR:

Grammar denotes syntactical rules in languages.

Types:-

| Grammar Type | Grammar Accepted | Language Accepted | Automation |
|---|---|---|---|
| Type0 | Unrestricted Grammar | Recursively Enumerable Language. | Turing Machine. |
| Type1 | Context Sensitive Grammar | Context Sensitive Language. | Linear Bounded Automata |
| Type2 | Context Free Grammar | Context Free Language | Pushdown Automata |
| Type3 | Regular (or) Grammar Regular Expression | Regular Language | Finite Automata |

## CONTEXT FREE GRAMMAR (CFG):-

Context Free Grammar (CFG),

$G = (V, T, P, S)$

Where, V = Set of Non-Terminals

T = Set of Terminals

P = Set of productions

S = Start Symbol.

# UNIT-III GRAMMARS AND APPLICATIONS

## CONTEXT FREE GRAMMAR & LANGUAGES:-

## DERIVATIONS & PARSE TREE:

Ex:1 Consider the grammar

$S \to aB / bA$

$A \to a / aS / bAA$

$B \to b / bS / aBB.$

write leftmost and rightmost derivations and draw parse tree for the string aababb.

**LMD:-**

$S \Rightarrow aB$

$\Rightarrow aaBB$

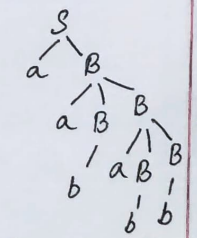
$\Rightarrow aabB$

$\Rightarrow aabaBB$

$\Rightarrow aababB$

$\Rightarrow aababb$

Parse Tree



**RMD:-**

$S \Rightarrow aB$

$\Rightarrow aaBB$

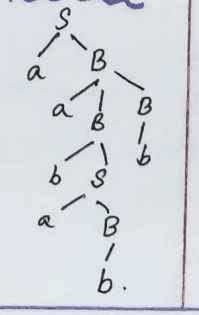
$\Rightarrow aaBaBB$

$\Rightarrow aaBaBb$

$\Rightarrow aaBabb$

$\Rightarrow aababb.$

Parse-Tree



Ex:2

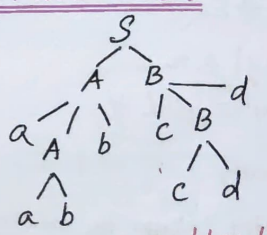
Consider the grammar G:

$S \to AB/\epsilon$, $A \to aAb/ab$

$B \to cBd/cd$, $C \to aCd/aDd$

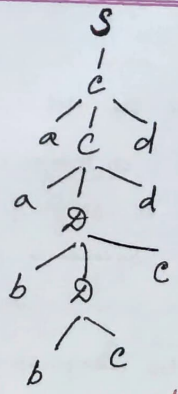
$D \to bDc/bC$. Show that the grammar G is Ambiguous for input string "aabbccdd".

**LMD PARSE TREE:**



$\Rightarrow$ aabbccdd //

**RMD PARSE TREE:**



$\Rightarrow$ aabbccdd //

Thus G is a Ambiguous Gramma

# SIMPLIFICATION OF CFG:

## * ELIMINATION OF ε-PRODUCTION:-

Ex:1 Eliminate ε-productions from the grammar.

$S \to AB$
$A \to aAA / \varepsilon$
$B \to bBB / \varepsilon$.

Sol:- S, A, B are nullable.

$S \to AB / B / A$
$A \to aAA / aA / a$
$B \to bBB / bB / b$.

Ex:2 Eliminate ε-productions from the grammar

$S \to XYZ$, $X \to 0X / \varepsilon$, $Y \to 1Y / \varepsilon$.

Sol:- X, Y are nullable

$S \to XYZ / YZ / XZ / Z$
$X \to 0X / 0$
$Y \to 1Y / 1$.

## * ELIMINATION OF UNIT PRODUCTION:-

Ex:1 Consider the grammar,

$E \to T / E+T$
$T \to F / T*F$
$F \to I / (E)$
$I \to a / b / Ia / Ib / I0 / I1$.

Sol:- Unit Productions are $E \to T$, $T \to F$, $F \to I$. After Eliminating Unit Productions

$E \to E+T / T*F / (E) / a / b / Ia / Ib / I0 / I1$
$T \to T*F / (E) / a / b / Ia / Ib / I0 / I1$
$F \to (E) / a / b / Ia / Ib / I0 / I1$.
$I \to a / b / Ia / Ib / I0 / I1$.

## * ELIMINATION OF USELESS SYMBOLS:-

Ex: Consider the grammar

$S \to aB / bX$
$A \to Bad / bSX / a$
$B \to aSB / bBX$
$X \to gBD / aBX / ad$.

Sol:-

Step1:- After removing non-generating symbols B,

$S \to bX$
$A \to bSX$
$X \to ad$

Step2:- After removing non-reachable symbol A,

$S \to bX$
$X \to ad$

## CHOMSKY NORMAL FORM

$NT \to Terminal$
$NT \to NT\ NT$

Ex: Construct a grammar in chomsky Normal form equivalent to the grammar,

$S \to bA / aB$, $A \to bAA / aS / a$,
$B \to aBB / bS / b$.

Sol:- $A \to a$, $B \to b$, already in CNF.

Let us take, $S \to bA / aB$,
It can be converted to,

$S \to C_a B$
$C_a \to a$
$S \to C_b A$
$C_b \to b$.

Let us take $A \to bAA / aS / a$
It can be converted to,

$A \to C_b D_1$
$D_1 \to AA$
$C_b \to b$
$A \to C_a S$
$C_a \to a$
$A \to a$

Let us take $B \to aBB / bS / b$,
It can be Converted to $\Rightarrow$

$B \to C_a D_2$, $D_2 \to BB$
$B \to C_b S$
$C_b \to b$
$B \to b$

# GREIBACH NORMAL FORM (GNF)

**Lemma : 1** — NT → One Terminal, any number of NTs.

If $A \to \alpha_1 B \alpha_2$ and $B \to \beta_1 / \beta_2 / \dots / \beta_r$ then

$A \to \alpha_1 \beta_1 \alpha_2 / \alpha_1 \beta_2 \alpha_2 / \alpha_1 \beta_3 \alpha_2 / \alpha_1 \beta_4 \alpha_2 / \dots / \alpha_1 \beta_r / \alpha_2$.

**Lemma : 2**

If $A \to A\alpha_1 / A\alpha_2 \dots$ and $A \to \beta_1 / \beta_2 / \dots \beta_r$ then

$A \to \beta_i$

$A \to \beta_i B$

$B \to \alpha_i$

$B \to \alpha_i B$.

Ex: Convert to GNF for the grammar $G = (\{A_1, A_2, A_3\}, \{a, b\}, P, A_1)$ where P consist of the following

$A_1 \to A_2 A_3$

$A_2 \to A_3 A_1 / b$

$A_3 \to A_1 A_2 / a$.

Sol: Let us consider, $A_3 \to A_3 A_1 A_3 A_2$

$A_3 \to b A_3 A_2 / a$

we can apply Lemma 2,

Now, $A_3 \to A_3 \underbrace{A_1 A_3 A_2}_{A \to A\alpha_1} / \underbrace{b A_3 A_2}_{\beta_1} / \underbrace{a}_{\beta_2}$

$A \to A\alpha_1$

$A \to \beta_1 / \beta_2$

$\Downarrow$

$A \to \beta_1, \; A \to \beta_2$

$A \to \beta_1 B, \; A \to \beta_2 B$.

$B \to \alpha_1$

$B \to \alpha_1 B$.

$A_3 \to A_3 A_1 A_3 A_2$

$A_3 \to b A_3 A_2 / a$

$\Downarrow$

$A_3 \to b A_3 A_2, \; A_3 \to a$

$A_3 \to b A_3 A_2 B, \; A_3 \to aB$

$B \to A_1 A_3 A_2$

$B \to A_1 A_3 A_2 B$.

Then we can apply Lemma now, we get,

$A_1 \to bA_3A_2BA_1A_3 / aBA_1A_3 / bA_3A_2A_1A_3 / aA_1A_3 / bA$

$A_2 \to bA_3A_2BA_1 / aBA_1 / bA_3A_2A_1 / aA_1 / b$

$A_3 \to bA_3A_2 / bA_3A_2B / aB / a$

$B \to bA_3A_2BA_1A_3A_3A_2 / aBA_1A_3A_3A_2 /$

$bA_3A_2A_1A_3A_3A_2 / aA_1A_3A_3A_2 / bA_3A_3A_2 /$

$bA_3A_2BA_1A_3A_3A_2B / aBA_1A_3A_3A_2B /$

$bA_3A_2A_1A_3A_3A_2B / aA_1A_3A_3A_2B / bA_3A_3A_2B$

## APPLICATIONS OF CONTEXT-FREE GRAMMARS

* Used in Parsing (Syntax Analysis) in Compiler Design.
* Natural Language Processing.
* Human Activities Recognition.

## CLOSURE PROPERTIES OF CONTEXT FREE LANGUAGES

* Union of two CFL's is context free
* Concatenation of two CFL's is context free.
* closure of a CFL is context free.
* Intersection of two CFL's is not context free.
* Intersection of a CFL and a regular language is context free.
* Complement of a CFL is not context free
* Substitution of a CFL is context free.
* Homomorphism of a CFL is context free.
* Inverse Homomorphism of a CFL is context free.

## PUMPING LEMMA FOR CFL :-

Let 'L' be a context free language then exists a constant 'n', such that if 'z' is any string in L, such that $|z| \geq n$, then we can write, $z = uvwxy$, subject to following conditions,

i) $vx \neq \varepsilon$

ii) $|vwx| \leq n$

iii) $uv^i wx^i y \in L \quad \forall i \geq 0.$

## PROBLEMS BASED ON PUMPING LEMMA :-

**Ex:1** Show that the language $L = \{0^n 1^n 2^n \mid n \geq 1\}$ is not context free.

**Sol:-** Assume that the given language is context free,

Let us take the string $z = 0^n 1^n 2^n$ where n is constant of pumping Lemma.

$z = uvwxy$, such that,

i) $vx \neq \varepsilon$, ii) $|vwx| \leq n$

iii) $uv^i wx^i y \in L$, $\forall i \geq 0$,

The string vwx cannot have all 3 symbols, 0's, 1's and 2's because $|vwx| \leq n$

case i) vwx has no 2's, $\therefore$ vx consists of only 0's & 1's. Put $i = 0$,

$uv^i wx^i y$, we get uwy,

$\therefore$ String uwy will have n no: of 2's but fewer than n 0's & 1's.

$\therefore uwy \notin L$

which is a Contradiction

$\therefore$ The given language is not context free.

**Ex:2** Show that the language $L = \{0^i 1^j 2^i 3^j \mid i \geq 1, j \geq 1\}$, is not context free.

**Sol:-**

Assume the given language is context free, let us take the string $z = 0^n 1^n 2^n 3^n$, where n is constant,

By Pumping Lemma,

$z = uvwxy$, such that

i) $vx \neq \varepsilon$,

ii) $|vwx| \leq n$

iii) $uv^i wx^i y \in L$, $\forall i \geq 0$,

The string vwx cannot involve all the symbols 0's 1's 2's & 3's. It can have atmost two symbols,

case i)

vwx consists of only symbol

eg. Assume vwx consists of 0's, put $i = 0$ in $uv^i wx^i y$, the string uwy will have n no: of 1's, 2's & 3's. But fewer than n 0's.

The number of 0's and 2's donot match

$\therefore uwy \notin L$ which is a Contradiction

$\therefore$ The given language is not Context free.

# PUSH DOWN AUTOMATA

**PDA Definition:**

PDA : $M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$

Design of PDA for the language $L = \{0^n 1^n / n \geq 1\}$ by final state.

**Solution:**

Nature of the Problem:

Number of 0's are equal to Number of 1's

Execution procedure:

$q_0$ – Accepting only one Zero

$q_1$ – Accepting more Zero's till 1

$q_2$ – Used to pop the stack i/p Symbol 0

$q_3$ – Accepting / Final State

**MOVE'S AND INSTANTANEOUS DESCRIPTIONS**

To design Moves:

$\delta(q_0, 0, Z_0) = (q_1, 0Z_0)$

$\delta(q_1, 0, 0) = (q_1, 00)$

$\delta(q_1, 1, 0) = (q_2, \varepsilon)$

$\delta(q_2, 1, 0) = (q_2, \varepsilon)$

$\delta(q_2, \varepsilon, Z_0) = (q_3, Z_0)$

Then PDA $M = \{Q, \Sigma, \Gamma, \delta, q_0, Z_0, q_3\}$

Where $Q = \{q_0, q_1, q_2, q_3\}$

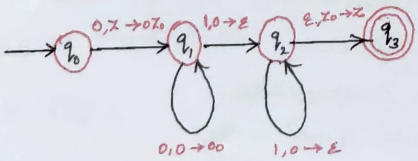
$\Sigma = \{0, 1\}$

$\Gamma = \{Z_0, 0\}$

$q_0$ is initial State

$Z_0$ is Stack Start Symbol

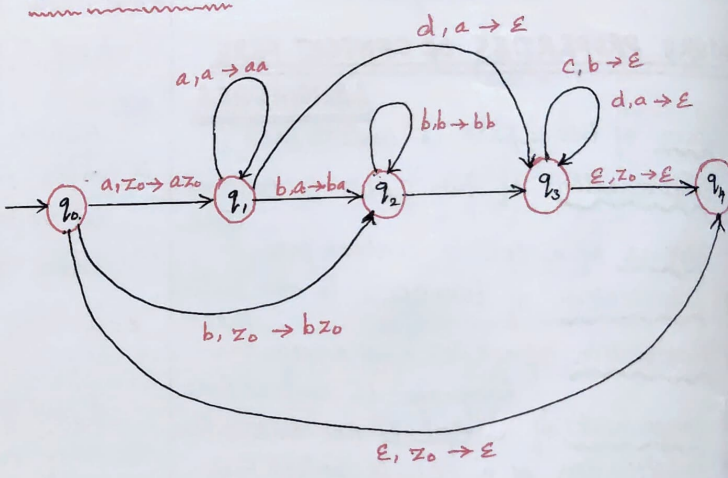
$q_3$ is Final State



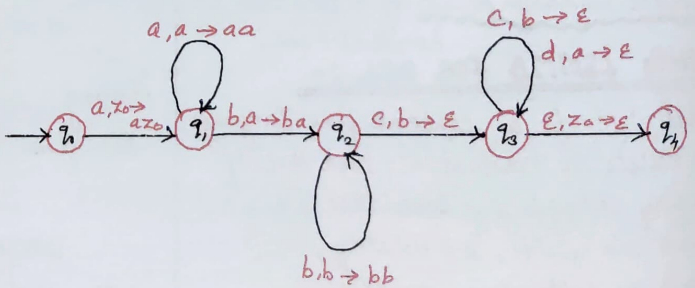

# THE LANGUAGE OF A PDA

Design of PDA for the language $L = \{a^n b^m c^m d^n\}$ $m, n \geq 0$ by Empty Stack. And also design for $m, n \geq 1$ by Empty Stack.

If $m, n \geq 0$.




If $m, n \geq 1$,



# Equivalance of PDA's AND CFG's

## PDA → CFG

Let $M = (\{q_0, q_1\}, \{0,1\}, \{X, Z_0\}, \delta, q_0, Z_0, \phi)$

Where.

$\delta$ is given by,

$\delta(q_0, 0, Z_0) = (q_0, XZ_0)$

$\delta(q_0, 0, X) = (q_0, XX)$

$\delta(q_0, 1, X) = (q_1, \varepsilon)$

$\delta(q_1, 1, X) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, X) = (q_1, \varepsilon)$

$\delta(q_1, \varepsilon, Z_0) = (q_1, \varepsilon)$

Construct CFG G generating N(M)

S productions are,

$P_1 : S \to [q_0, Z_0, q_0]$

$P_2 : S \to [q_0, Z_0, q_1]$

$P_3$ : Let us take $\delta(q_0, 0 Z_0) = (q_0, XZ_0)$

Productions are :

$P_3 : [q_0, Z_0, q_0] \to 0\,[q_0, X, q_0]\,[q_0, Z_0, q_0]$

$P_4 : [q_0, Z_0, q_1] \to 0\,[q_0, X, q_1]\,[q_1, Z_0, q_1]$

$P_5 : [q_0, Z_0, q_1] \to 0\,[q_0, X, q_0]\,[q_0, Z_0, q_1]$

$P_6 : [q_0, Z_0, q_1] \to 0\,[q_0, X, q_1]\,[q_1, Z_0, q_1]$

Let us take,

$\delta(q_0, 0, X) = (q_0, XX)$

Productions are

$P_7 : [q_0, X, q_0] \to 0\,[q_0, X, q_0]\,[q_0, X, q_0]$

$P_8 : [q_0, X, q_0] \to 0\,[q_0, X, q_0]\,[q_1, X, q_0]$

$P_9 : [q_0, X, q_1] \to 0\,[q_0, X, q_0]\,[q_0, X, q_1]$

$P_{10} : [q_0, X, q_1] \to 0\,[q_0, X, q_1]\,[q_1, X, q_1]$

$P_{11} : [q_0, X, q_1] \to \varepsilon$ $\quad \delta(q_0, 1, X) = (q_1, \varepsilon)$

$P_{12} : [q_1, Z_0, q_1] \to \varepsilon$ for $\quad \delta(q_1, \varepsilon, Z_0) = (q_1, \varepsilon)$

$P_{13} : [q_1, X, q_1] \to \varepsilon$ for $\quad \delta(q_1, \varepsilon, X) = (q_1, \varepsilon)$

$P_{14} : [q_1, X_1, q_1] \to$ for $\quad \delta(q_1, 1, X) = (q_1, \varepsilon)$

$P_2$, $P_6$, $P_{10}$, $P_{12}$, $P_{13}$, $P_{14}$ are the only productions that can able to produce the terminals. So we have to delete the other productions

## CFG → PDA

Construct the given Expression to a PDA

$E \to I \mid E*E \mid E+E \mid (E)$

$I \to a \mid b \mid I_a \mid I_b \mid I_0 \mid I_1$

PDA is given by.

$M = (\{q\}, \{+, *, a, b, C, ), 0, 1\}, \{I, E, +, *, a, b, (, ), 0, 1\}, \delta, q, E)$

Where $\delta$ is defined by,

(i) $\delta(q_1, \varepsilon, I) = \{(q,a), (a,b), (q, I_a), (q, I_b), (q, I_0), (q, I_1)\}$

(ii) $\delta(q, \varepsilon, E) = \{(q, I), (q, E+E), (q, E*E), (q, (E))\}$

(iii)

$\delta(q, a, a) = (q, \varepsilon)$ $\delta(q, c, C) = (q, \varepsilon)$

$\delta(q, b, b) = (q, \varepsilon)$ $\delta(q, ), )) = (q, \varepsilon)$

$\delta(q, 0, 0) = (q, \varepsilon)$ $\delta(q, +, +) = (q, \varepsilon)$

$\delta(q, 1, 1) = (q, \varepsilon)$ $\delta(q, *, *) = (q, \varepsilon)$

## Programming Techniques for Turing Machine

1. storage in Finite control
2. Multiple Track
3. checking off symbols
4. Subroutine.

## storage in Finite Control

TM has finite Control

It store the following Information

1. Current state
2. Current Symbol

$\delta(q_0, a) = (q_1, b, R)$

current state, Input, next state, Modified Symbol, Movement Direction

So. Tm can recognise the language $0^n 1^n$

## 2. Multiple Tracks

Using multiple track, we can perform subtraction

| # | 1 | 1 | 1 | 1 | 1 | # |
|---|---|---|---|---|---|---|
| B | B | B | B | 1 | 1 | B |
| B | 1 | 1 | 1 | B | B | B |

11111 − 11 = 111

## Checking off symbols

* Tm can recognize any type of string using off symbols.
* Using off symbols, it decide the direction left or right.

→ Tm can recognize the

$L = \{ wcw \}$

$w = aba \in aba$

↓

off symbols

The Tm decide the direction depend upon the 'ε' off Symbol.

## Subroutine

* We can write subroutine as a TM
* We can construct subroutine for the following

$f(a, b) = a + b$.

## Turing Machine [TM]

Design Tm to recognize

$L = 0^n 1^n$

n=3

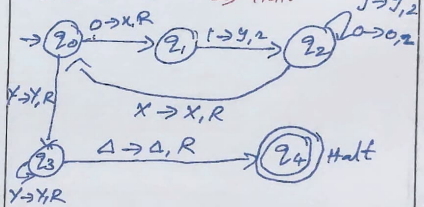
0 0 0 1 1 1

X 0 0 1 1 1

X 0 0 Y

X X 0 Y Y

X X X Y Y Y → Halt




### Transition configuration

1. $\delta(q_0, 0) = (q_1, X, R)$
2. $\delta(q_1, 0) = (q_1, 0, R)$
3. $\delta(q_1, 1) = (q_2, Y, L)$
4. $\delta(q_2, 0) = (q_2, 0, L)$
5. $\delta(q_2, X) = (q_0, X, R)$
6. $\delta(q_0, Y) = (q_3, Y, R)$
7. $\delta(q_3, \Delta) = (q_4, \Delta, R)$

## Comparision of PDA and TM

### Finite Automata

1. It recognizes Regular language
2. The i/p Tape is of finite length
3. one direction movement

### PDA

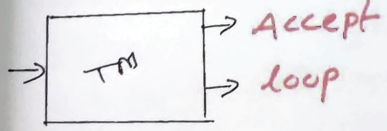
1. It will recognize CFL, RL
2. stack memory used
3. Push & Pop operation

### TM

1. It recognize ALL language
2. Infinite length tape is used.
3. Head can move in both directions.

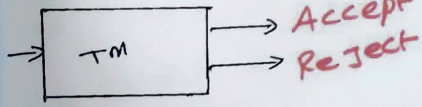
# Undecidability

## Recursive Enumerable language




* R.E language can be accepted or recognize by TM.
* TM will not enter into rejecting state, it means TM can loop forever.

## Recursive language




The recursive language can be decided by TM, which means it will enter into accept state or reject state for the string of language.

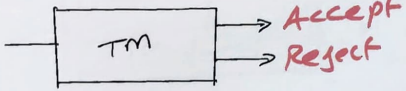
## Undecidable problem

$Halt_{TM}$ is undecidable

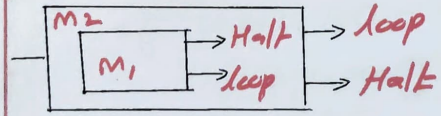
we can prove that $Halt_{TM}$ is undecidable.

## Basic idea

Pf ($Halt_{TM}$ is decidable)
{ Decidable language is also Turing acceptable
}




But Halting problem is undecidable.




[If $M_1$ Halt Then
$M_2$ will loop
If $M_1$ in loop Then
$M_2$ will Halt]

* It is contradiction
* So $Halt_{TM}$ is undecidable.

* Prove that the diagonalization language $L_d$ is not Recursively Enumerable $L_d$.

creation of Diagonalization lang.

| | | | |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 |
| 1 | 1 | 1 | 0 |
| 1 | 0 | 1 | 1 |

Dragonal value = 0111
complement = 1000

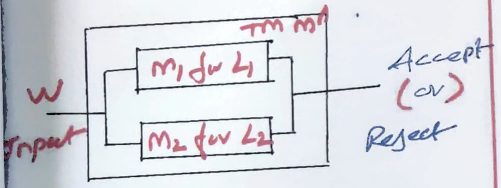
[If w is in $L_d$ Then
TM will accept 'w'
But By the definition of $L_d$
$w \notin L_d$]

* Thus $L_d$ is not recursively Enumerable language.

## Recursive & Recursively Enumerable languages

* **Theorem**

If $L_1$ & $L_2$ are recursive language then $L_1 \cup L_2$ also recursive language.

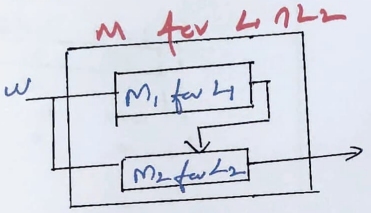



[If $w \in (L_1 \cup L_2)$ Then
$M_1$ Halt or $M_2$ Halt
If $w \notin (L_1 \cup L_2)$ Then
$M_1$ & $M_2$ will not Halt]

* Hence $L_1$ & $L_2$ are recursive then $L_1 \cup L_2$ also recursive language.
* we conclude that M' behaves for the language of $[L_1 \cup L_2]$.

## Theorem

If two language $L_1$ and $L_2$ are recursively Enumerable then their intersection $L_1 \cap L_2$ also recursive Enumerable.

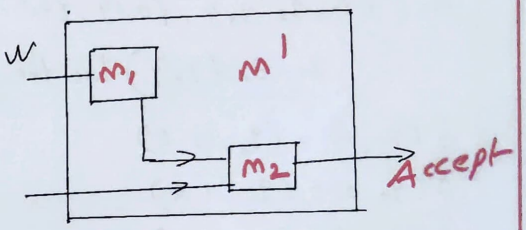



[If $w \in L_1$ & $w \in L_2$ Then
$w \in L_1 \cap L_2$
If $w \in L_1$ [or] $w \in L_2$ Then
$w \notin L_1 \cap L_2$]

* Thus $L_1 \cap L_2$ is Recursively Enumerable language
* we conclude that M behaves for the language of $L_1 \cap L_2$.

## Rice Theorem

Every non trivial property of Recursively Enumerable language is undecidable.




1. [If $w \in M_1$ [or] $w \notin M_1$]
2. [If $M_1$ accept w Then M' accept the language of $M_2$]

* By the condition of 1 & 2 we can not decide code for M'.
* This proves that the property of Recursively Enumerable is undecidable.

# UNDECIDABILITY - 3

## Enumerating Binary Code

Obtain the code for $\langle M, 1011 \rangle$ where

$M = (\{q_1, q_2, q_3\}, \{0,1\}, \{0,1,B\}, \delta, q_1, B, \{q_2\})$ has the moves

$\delta(q_1, 1) = (q_3, 0, R)$

$\delta(q_3, 0) = (q_1, 1, R)$

$\delta(q_3, 1) = (q_2, 0, R)$

$\delta(q_3, B) = (q_3, 1, L)$

Consider the following replacements

$q_1$ by one zero — left by one zero

$q_2$ by two zeros — right by 2 zeros (directions)

$q_3$ by 3 zeros (stats)

0 by one zero

1 by 2 zeros

B by 3 zeros (Tape Symbols)

code for $\langle M, 1011 \rangle$ is

111 0100100010100 11 000101001100
(Code-1) (Code-2)

11 000100100101000 11 000010001000100010
(Code-3) (Code-4)

111 1011
w

## Post's Correspondence Problem

Let $\Sigma = \{0,1\}$

Let A and B be strings. Find the instance of Post's correspondence problem

| i | List A: $w_i$ | List B: $x_i$ |
|---|---|---|
| 1 | 1 | 111 |
| 2 | 10111 | 10 |
| 3 | 10 | 0 |

Let $M = A$,

$i_1 = 2$, $i_2 = 1$, $i_3 = 1$, $i_4 = 3$

Take this combination 2113

By concatinating strings in this series

$w_2 w_1 w_1 w_3 = x_2 x_1 x_1 x_3$

10111 1 1 10 = 10 111 111 0

∴ Instance of PCP is 2113

For another instance 2113 2113 of a PCP has a solution.